

Adenoviruses

**Background document for the
WHO Guidelines for drinking-water quality and the
WHO Guidelines on sanitation and health**

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Highlights

- There is strong evidence linking transmission of human adenoviruses (HAdVs) with inadequate sanitation and poor hygiene. HAdVs are important waterborne pathogens. However, outbreaks are primarily associated with recreational waters. In view of their prevalence and detection in water, drinking-water represents a likely but unconfirmed source of infection. HAdVs are also considered important in the context of drinking-water because of their significant resistance to UV light.
- Adenoviruses are generally only pathogenic to the species of origin, including HAdVs. However, it has been suggested that they may cross species barriers.
- Transmission occurs by the faecal–oral and oral–oral routes and by hand–eye contact. Person-to-person contact plays a major role in transmission, but inhalation of aerosolized particles, ingestion of contaminated recreational waters, or fomites can also play a role. The consumption of contaminated food or drinking-water may be an important source of enteric illness, although there is limited evidence supporting these routes of transmission.
- Most infections are asymptomatic, but one third of all the known adenovirus types cause diseases ranging from gastroenteritis, respiratory infections, eye infections, acute haemorrhagic cystitis, and meningoencephalitis.
- Adenoviruses infect people worldwide, and infections occur year-round, without marked seasonality. Adenovirus infections are particularly important in young children (less than 2 years of age), the elderly and immunosuppressed adults.
- The relative infectivity of adenoviruses is high. The ID₅₀ (50% infective dose) could be as low as 0.74 virus particles.
- Non-standard culture-based and molecular methods for detection of adenoviruses in environmental samples are available.
- Adenoviruses have been detected in drinking-water, raw sewage, reused or reclaimed wastewater, soil, sediment and seawater. HAdVs can persist for long periods of time in water.
- Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including from adenoviruses. However, as they spread easily from person-to-person other control measures, including adequate hygiene, are required.
- Reduction of HAdVs by wastewater and drinking-water treatment varies depending on the process. Chlorination is effective, while UV light is ineffective for inactivating HAdVs.
- Due to the smaller size of HAdVs relative to *E. coli*, resulting in lower removal by treatment processes such as filtration, and their greater environmental persistence, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the absence of HAdVs in drinking-water. However, the presence of *E. coli* is a suitable indicator of the possible presence of HAdVs.

1 General description

The family Adenoviridae is classified into five recognized genera: *Mastadenovirus* with mammalian hosts, *Aviadenovirus* with avian hosts, and *Atadenovirus*, *Ichtadenovirus* and *Siadenovirus* with a broad range of nonhuman hosts (Davison et al., 2003; ICTV, 2018). Adenoviruses consist of a double-stranded, linear DNA genome in a non-enveloped icosahedral capsid with a diameter of about 90 nm to 100 nm (Jiang, 2006; Allard & Vantarakis, 2017).

Human adenoviruses (HAdVs) have been classified into seven species (A to G) on the basis of their physical, chemical and biological properties (Clark & McKendrick, 2004; Allard & Vantarakis, 2017). These species vary in their target sites of infection, with A, D and F targeting the gastrointestinal tract (Jiang, 2006). HAdVs are further classified into 68 distinct types, identified using traditional serotyping (first 51 types) and molecular genotyping methods (Clark & McKendrick, 2004; Allard & Vantarakis, 2017; Cook & Radke, 2017; Kim et al., 2017). Adenoviruses are generally only pathogenic to the species of origin. However, they may cross the species barrier due to recombination of different types (Allard & Vantarakis, 2017; Borkenhagen et al., 2019).

2 Human health effects

HAdVs cause a wide range of infections with a spectrum of clinical manifestations. These include infections of the gastrointestinal tract (gastroenteritis), the respiratory tract (acute respiratory diseases, pneumonia, pharyngoconjunctival fever), the urinary tract (cervicitis, urethritis, haemorrhagic cystitis), the eyes (epidemic keratoconjunctivitis, also known as “shipyard eye”; pharyngoconjunctival fever, also known as “swimming pool conjunctivitis”) and the meninges (meningoencephalitis) (Webber, 2005; Jiang, 2006; Shieh, 2022). Different serotypes are associated with specific illnesses; for example, gastrointestinal symptoms are most commonly caused by types 40 and 41 (species F), but have also been observed from infection with type 52 (species G) and some types of species D (Allard & Vantarakis, 2017), whereas respiratory tract infections are more commonly associated with species B (types 3, 7, 16 and 21), E (type 4) and C. Asymptomatic infection also occurs (Zhang et al., 2011) and accounts for two thirds of all HAdV infections (Allard & Vantarakis, 2017).

Young children, the elderly and immunosuppressed adults are more susceptible to adenoviruses and are more likely to have severe infections (Kotloff et al., 2013; Platts-Mills et al., 2015; Allard & Vantarakis, 2017; Cook & Radke, 2017). Infection in otherwise healthy adults is rare. Mortality rates are higher among sensitive populations, especially persons with rare congenital immunodeficiency and those with acquired immunodeficiency resulting from underlying conditions, such as immunosuppressive therapy, use of corticosteroids, radiation therapy, severe burns, or AIDS (Echavarría, 2008).

3 Epidemiology

Adenovirus infections have been detected worldwide, and infections occur year-round without marked seasonality (Hundesha et al., 2006; Okoh et al., 2010; Hewitt et al., 2013). In low-income settings, adenovirus types 40 and 41 have been identified as an important cause of diarrhoeal

disease among children less than 2 years of age (Kotloff et al., 2013; Platts-Mills et al., 2015; Lee et al., 2020) and are estimated to be associated with significant diarrhoeal disease burden among children under five (GBD, 2020).

4 Transmission

HAdVs are transmitted by the faecal-oral and oral-oral routes and by hand-eye contact (Allard & Vantarakis, 2017). Person-to-person contact plays a major role in transmission, but inhalation of aerosolized particles and ingesting contaminated recreational waters and fomites can also play a role. The consumption of contaminated food or drinking-water may be an important source of enteric illness, although there is no substantial evidence supporting these routes of transmission. Adenoviruses are generally only pathogenic to the species of origin but there is some evidence of zoonotic transmission (Allard & Vantarakis, 2017; Borkenhagen et al., 2019).

There have been numerous outbreaks associated with hospitals, military establishments, child-care centres and schools (Saint-Pierre Contreras et al., 2023). Many outbreaks have also been associated with recreational water sites (Jiang, 2006; Sinclair et al., 2009). Contamination of recreational waters with adenoviruses has been well-documented in areas receiving even properly treated wastewater (Okoh et al., 2010). Secondary transmission of adenoviruses following infection from recreational waters may be substantial. Symptoms recorded in most outbreaks were acute respiratory disease, keratoconjunctivitis and conjunctivitis. Outbreaks of gastroenteritis have also been reported.

In a dose-response model by Teunis et al. (2016), based on a study originally conducted by Couch et al. (1963), the mean ID₅₀ (50% infective dose, i.e. the number of organisms required to infect 50% of exposed subjects) of HAdVs by ingestion was calculated to be 0.74 virus particles. Thus, the relative infectivity is considered high¹. The incubation period is typically 2 to 14 days after exposure, depending on the virus serotype (Allard & Vantarakis, 2017). Individuals infected with enteric types of adenoviruses may shed up to 10¹¹ particles/gram of faeces or about eight times lower if infected with non-enteric adenoviruses (Hewitt et al., 2013; Allard & Vantarakis, 2017). Shedding can continue for months after symptoms resolve. For adenovirus types affecting the respiratory tract, high levels of viral particles are also found in sputum or oral secretions (Allard & Vantarakis, 2017). Serotype-specific immunity is conferred following infection with enteric adenoviruses, but this wanes over time (Allard & Vantarakis, 2017).

5 Detection methods

Non-standard culture-based, as well as molecular-based methods are available for HAdVs (Chapron et al., 2000; Jiang, 2006; Polston et al., 2014; Corpuz et al., 2020). Adenoviruses can be cultivated on numerous cell types; however, the efficiency of replication tissue or cell cultures varies with serotypes and a lengthy incubation period is required (Jiang, 2006). In

¹ From experiments with human volunteers, from epidemiological evidence and from experimental animal studies. Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: High means infective doses can be 1–10² organisms or particles, Moderate 10²–10⁴ and Low > 10⁴.

addition, serotypes 40 and 41, most commonly associated with gastroenteritis, grow very slowly, may not produce cytopathic effects on most cell lines, and thus may be overgrown by other faster growing viruses, especially enteroviruses (Jiang, 2006). Molecular-based techniques are faster and more sensitive to detect adenoviruses in environmental samples, including raw sewage, treated wastewater, surface waters, and drinking-water (Chapron et al., 2000; Jiang, 2006). However, distinguishing between viable and non-viable viruses is a shortcoming of molecular methods. Combining cell culture infectivity and molecular methods make it possible to improve the detection of HAdVs, including those that do not produce visible cytopathogenic effects in host cells (Rodriguez et al., 2013).

6 Source and occurrence

HAdVs have been widely detected in environmental matrices affected by human sewage, including raw water sources and treated drinking-water supplies (Jiang, 2006; Shi et al., 2021). In a systematic review and meta-analysis carried out by Takuissu et al. (2023), the global prevalence rate of adenoviruses (human and animal) in various water matrices was 59.3%. Adenoviruses were detected in various waters around the world. Prevalence rates in specific matrices are provided in the specific sub-sections below. The global prevalence rate was significantly different among countries with higher rates in Uganda (94.3%), Sweden (90.7%) and Germany (89.8%).

The presence of adenoviruses in environmental media has not been found to correlate with the presence of faecal indicator bacteria (Allard & Vantarakis, 2017). Given their specificity to humans and prevalence and persistence in drinking-water sources, sludge and wastewater, HAdVs have been proposed as a viral marker of human faecal pollution in water bodies (Hewitt et al., 2013; Teixeira et al., 2020; Garcia et al., 2022).

6.1 Faecal pollution sources

Adenoviruses have been detected frequently in raw sewage (in up to 100% of samples) and in high numbers (up to 10^{10} GC/L) (Allard & Vantarakis, 2017; Corpuz et al., 2020). Similarly, high frequencies and concentrations of HAdVs have been reported in treated wastewater due to their high resistance to treatment processes (La Rosa et al., 2010; Qiu et al., 2015; Allard & Vantarakis, 2017; Corpuz et al., 2020; Shi et al., 2021). In a systematic review and meta-analysis (Takuissu et al., 2023), the prevalence rates of adenoviruses in samples of untreated ($n=3690$ from 81 studies) and treated ($n=2657$ from 57 studies) wastewater were 83.2% and 75.4%, respectively. A systematic review on the presence of adenovirus in sewage sludges reported concentrations ranging from 10^4 to 10^5 GC/L in primary sludge to 10^5 GC/mL in activated sludge and 2.5×10^3 GC/g dry weight in digested sludge (Gholipour et al., 2022a). Adenoviruses can survive for months in biosolids with no degradation (Allard & Vantarakis, 2017).

6.2 Surface water

Adenoviruses have been detected in surface water worldwide in concentrations up to 10^5 GC/L (Allard & Vantarakis, 2017; Bailey et al., 2021; Shi et al., 2021). In a systematic review and meta-analysis (Takuissu et al., 2023), the prevalence rate of adenoviruses in samples of surface water ($n=10\ 920$) from 135 studies was 49.5%. Studies suggest that protists in surface waters

may act as reservoirs or vehicles of transmission for adenoviruses by internalizing the virus particles (Allard & Vantarakis, 2017).

HAdVs persist for long periods in surface water,² with persistence highly dependent on temperature. The time taken for a 1 log₁₀ reduction in a study of serotype 41 infectivity in surface water was 26 days at 4 °C and 3.2 days at 37 °C (Prevost et al., 2016). In this study, infectivity persisted for more than 70 days at 4 °C and 20 °C and more than 25 days at 37 °C.

6.3 Groundwater

In a systematic review and meta-analysis (Takuissu et al., 2023), the prevalence rate of adenoviruses in groundwater samples (n=2525) from 19 studies was 18.5%.

HAdVs persist for long periods in groundwater (Kotwal & Cannon, 2014). In groundwater at 4 °C, infectivity reductions of Adenovirus 2 were estimated as 0.0076 log₁₀ per day, whereas at 20 °C, the reduction was 0.0279 log₁₀ per day (Ogorzaly et al., 2010).

6.4 Drinking-water

In tap water, HAdVs have been shown to be prevalent and occur in substantial numbers of up to 10⁷ GC/L (Cho et al., 2000; Albinana-Gimenez et al., 2006; Allard & Vantarakis, 2017). In a systematic review and meta-analysis (Takuissu et al., 2023), the prevalence rate of adenoviruses in samples of drinking-water (n=1511) from 22 studies was 22.7%. Notably, HAdVs have been detected in treated drinking-water supplies that met accepted specifications based on conventional indicator organisms (Grabow et al., 2001; Jiang, 2006; Shi et al., 2021). In Nagpur, India, Bivins et al. (2021) detected an increase in the proportion of samples positive for adenoviruses A to F by molecular methods at household taps served by an intermittent water supply compared to those served by a continuous water supply. Rashid et al. (2021) detected HAdVs in 20% of tap water samples in Karachi, Pakistan, while Shi et al. (2021) detected HAdVs in 100% of tap water samples in Tianjin City, China.

HAdVs persist for long periods in drinking-water, and have been shown to survive for up to one year (Rzežutka & Cook, 2004; Kotwal & Cannon, 2014). Temperature modifies the survival of adenoviruses in drinking-water: in colder temperatures, survival can be up to 1 year, while at room temperature, it may be as long as 2 to 3 months (Rzežutka & Cook, 2004). In a study of Adenovirus 41 infectivity in drinking-water, the time taken for a 1 log₁₀ reduction was 19 days at 4 °C and 2.5 days at 37 °C (Prevost et al., 2016).

6.5 Seawater

The presence of adenoviruses in seawater has been confirmed by many studies (Jiang et al., 2001; Fong et al., 2005; Allard & Vantarakis, 2017).

² Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: Detection period for infective stage in water at 20 °C: Short, up to 1 week; Moderate, 1 week to 1 month; Long, over 1 month.

6.6 Soil and crops

Both soil and crops can be contaminated with enteric viruses through irrigation with contaminated water or the application of organic waste (Gholipour et al., 2022b). In wastewater-irrigated crops, adenoviruses have been detected frequently in low-income settings (Silverman et al., 2013), and there have also been many reports of detection in shellfish, even when they meet standards for bacterial indicators (Allard & Vantarakis, 2017).

6.7 Surfaces

Adenoviruses have been found on fomites, e.g. shared utensils (Rzežutka & Cook, 2004; LaRosa et al., 2013; Allard & Vantarakis, 2017).

7 Significance of sanitation

Safe sanitation systems are important barriers against the spread of adenoviruses (Allard & Vantarakis, 2017). However, given the importance of person-to-person, oral-oral and hand-eye transmission, additional control measures, including adequate hygiene, are needed to effectively prevent transmission.

8 Significance in drinking-water

Adenoviruses are important waterborne pathogens. Many HAdV outbreaks have been associated with recreational waters (Jiang, 2006; Sinclair et al., 2009). HAdVs have been shown to occur in substantial numbers in raw water sources and treated drinking-water and the latter represents a likely but unconfirmed source of HAdV infections. Adenoviruses are also considered important in the context of drinking water because of their significant resistance to inactivation by UV light (see section 9.2).

9 Prevention and management

Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains, are needed to manage microbial risks, including from adenoviruses.

The effectiveness of sanitation and drinking-water treatment systems depends on the selection and operation of technologies and the characteristics of the target organisms. Treatment processes based on size exclusion (e.g. filtration through media or soil in septic tank drainage areas) or settling in primary wastewater treatment processes will be limited for removing viruses, including adenoviruses. Adenoviruses have low resistance to oxidizing chemical disinfectants such as chlorine but are highly resistant to disinfection by UV light.

9.1 Sanitation systems

Sanitation safety plans (WHO, 2022) should include control measures to safely collect, treat and discharge or reuse wastewater and faecal sludges. Removal or inactivation of adenoviruses

by wastewater treatment processes is variable, although the data are limited (Allard & Vantarakis, 2017). Based on available evidence, the reduction of adenoviruses by treatment processes is likely to be lower than the reduction of faecal indicator bacteria such as *E. coli*.

There are no data reported on removal of adenoviruses by on-site systems. In centralized sewage treatment systems, the removal of adenoviruses in primary clarifiers is negligible (Kuo et al., 2010). Reported removals of adenoviruses by other processes range from 1 to 2 log₁₀ for wetlands, negligible to 1.2 log₁₀ for stabilization ponds, 1.0 to 2.3 log₁₀ for activated sludge, and greater than 4 log₁₀ for membrane bioreactors (Rodriguez-Manzano et al., 2012; Verbyla & Mihelcic, 2015; Allard & Vantarakis, 2017).

Adenoviruses exhibit strong adsorption properties to sewage sludge (Yin et al., 2018) and have been found to be more strongly associated with sludge and suspended solids than other viruses in anaerobic reactors (Verbyla, 2015). Multiple strains of adenoviruses (including respiratory strains) have been detected in sewage sludge at concentrations of 10² to 10⁷ GC/g, and some adenoviruses are likely to remain viable in the sludge (Albinana-Gimenez et al. 2006; Bofill-Mas et al. 2006; Gholipour et al., 2022a, Sidhu & Toze 2009; Viau & Peccia, 2009; Viau et al., 2011; Wong et al. 2010; Bibby & Peccia, 2013).

Since potentially viable adenoviruses may be present in sewage sludge, faecal sludge, and septage, attention must be given to solids management at on-site and centralized facilities. In biosolids, a log₁₀ reduction of adenovirus concentrations required 12 days at 20 °C and 29 days at 4 °C (Wei et al., 2009). Infectious adenoviruses were detected in dewatered biosolids and in biosolids treated by mesophilic anaerobic digestion (Wong et al., 2010). Treatment with slaked lime to achieve a pH of 12 reduced adenovirus concentrations in sludge from 10⁴ per mL to undetectable levels after only 24 hours (Hansen et al., 2007). In faecal sludge adjusted to pH 9, a log₁₀ reduction of adenoviruses was observed in 3 to 4 days at 28 °C and in 25 days at 10 °C; when the pH was 7.7 to 7.8, 56 days at 33 °C and 214 days at 17 °C were required to achieve a log₁₀ reduction (Magri et al., 2015).

9.2 Drinking-water supplies

Within a water safety plan (WHO, 2012; 2023), control measures to manage potential risks from HAdVs should include protection of source waters (surface or groundwater) from contamination by human faecal wastes, application of adequate treatment processes and protection of drinking-water quality during distribution and storage.

There are limited data on removal of HAdVs by conventional drinking-water treatment processes (coagulation, flocculation, sedimentation and filtration). Bench- and pilot-scale studies suggest that adenoviruses are removed equally or more effectively than other enteric viruses or bacteriophages by conventional treatment (Abbaszadegan et al., 2007; Bauer et al., 2011; Shirasaki et al., 2018). Adenoviruses have low resistance³ to chlorine, Ct (product of

³ Within pathogen species and groups, there are likely to be variations in resistance, which could be further impacted by characteristics of the water supply and operating conditions. Resistance is based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: 99% inactivation at 20 °C where, generally, Low represents a Ct₉₉ of < 1 min.mg/L, Moderate 1–30 min.mg/L and High > 30 min.mg/L (where C = the concentration of free chlorine in mg/L and t = contact time in minutes) under the following conditions: the infective stage is freely suspended in water treated at conventional doses and contact times, and the pH is between 7 and 8.

disinfectant concentration and contact time) values for 2 log₁₀ inactivation at 5 °C and pH 8 for HAdV types 2, 40, and 41 were less than 0.04 mg.min/L (Cromeans et al., 2010). While adenoviruses are susceptible to chlorine, some are relatively resistant to monochloramine. For example, the Ct values recorded for HAdV 2 were 600 and 990 mg.min/L for a 2 log₁₀ reduction at pH 7 and pH 8, respectively (Cromeans et al., 2010). Adenoviruses are among the most resistant waterborne pathogens to UV light; the doses needed to achieve 2 log₁₀ inactivation of types 2, 15, 40 and 41 are up to 83 mJ/cm² to 110 mJ/cm² (Hijnen et al., 2006).

Due to the smaller size of HAdVs relative to *E. coli*, resulting in lower removal by treatment processes such as filtration, and their greater environmental persistence, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the absence of HAdVs in drinking-water. However, the presence of *E. coli* is a suitable indicator of the possible presence of HAdVs.

10 ICD-11 classification

XN000	Adenovirus
1A20	Enteritis due to Adenovirus
1C8E.2	Meningitis due to Adenovirus
1D84.0	Conjunctivitis due to Adenovirus
CA02.10	Pharyngitis due to Adenovirus
CA40.10	Pneumonia due to Adenovirus
1D84.Y	Other specified viral conjunctivitis, adenoviral pharyngoconjunctivitis
1D90	Adenovirus infection of unspecified site

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Annex. Approach to content development and declarations of interest

The background documents were developed drawing from the following resources (i) microbial fact sheets from the fourth edition of the WHO Guidelines for Drinking-Water Quality (ii) pathogen-specific chapters in the Global Water Pathogen Project (GWPP) book “*Sanitation and Disease in the 21st Century*”, an update of the seminal book by Feachem et al. 1983 “*Sanitation and Disease: Health Aspects of Excreta and Wastewater Management*” and (iii) a rapid review of the literature at the start of the development of the background documents and before finalization in December 2023, focusing on identifying additional recent high quality literature reviews on a given pathogen. The documents were prepared by members of the WHO drinking-water quality microbiology group as well as external scientists with expertise in the relevant pathogens. The documents went through several iterations based on inputs from members of the WHO drinking-water quality microbiology working group and WHO technical staff. In addition, each document was reviewed by at least two external peer reviewers including authors of the GWPP book chapters. The extensive review process was designed to ensure technical accuracy and to identify any data gaps. The background documents were discussed at three expert meetings to resolve any issues and ensure the content reflected a consensus of expert opinions. Upon satisfactory resolution of any further data gaps or issues, the background documents went through technical editing before finalization.

Before the key expert meetings mentioned above, and for this work more broadly, all microbiology working group members submitted declarations of interest to WHO. These were to disclose any potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of the guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meetings or covered by the guidance.